EVIDENCE BRIEF · SAMPLE

GLP-1 weight-loss drugs: what the research says about heart risk and death

For: life and health insurance product, underwriting and claims teams · Date: 6 Nov 2026 · Basis: paper abstracts only (see page 2)

Bottom line. In randomised trials of overweight adults without diabetes, GLP-1 and tirzepatide users had about 20% lower odds of heart events and of death from any cause than placebo users. Trials are short and select healthy volunteers, so this does not tell us lifetime mortality for people who stay on the drug, or for those who stop it. Known side effects are mostly digestive. A few rarer ones (thyroid cancer, eye problems, anaesthesia risk) are still being studied.

1. What the evidence shows

QUESTION	FINDING	SOURCE	STRENGTH
Heart events (MACE), adults without diabetes	Odds ratio 0.79 (95% CI 0.71 to 0.89) vs placebo	Meta-analysis, 16 trials, 28,168 people (Stefanou 2024)	Strong: pooled randomised trials
Death from any cause	Odds ratio 0.80 (0.70 to 0.92)	Same	Strong, but short follow-up
Death from heart causes	Odds ratio 0.84 (0.71 to 1.01): a trend, not clearly different from no effect	Same	Moderate
Heart attack	Odds ratio 0.72 (0.61 to 0.86). Stroke: no clear link	Same	Strong for heart attack; stroke data limited
Semaglutide, heart events, people with diabetes or obesity	Risk ratio 0.71 reported for semaglutide (confidence range not visible in the abstract)	Meta-analysis, Zhou 2026	Moderate: only the headline number was read
Heart failure	Lower risk in obese patients with heart or kidney disease, but not in non-obese ones	Meta-analysis, Zhang 2026	Moderate

2. Safety signals that matter for claims

SIGNAL	WHAT THE AUTHORS SAY	SOURCE
Pancreatitis, pancreatic cancer	Earlier concern "dispelled by long-term clinical trials"	Jalleh 2026 (JCI review)
Thyroid cancer	"May confer an increased risk"; still open	Jalleh 2026; Ueda 2026 lists it among seven serious events reviewed
Eye complications	Can follow rapid blood-sugar drops; screening before starting may avoid them	Jalleh 2026
Anaesthesia and endoscopy	Slower stomach emptying raises the chance of food remaining in the stomach, so aspiration risk	Jalleh 2026
Staying on the drug	Side effects affect how long people keep taking it	Jalleh 2026

3. What this could mean for a risk team (my judgment, not a finding)

- **Mortality credit:** a case for some credit is plausible (about 55% in my judgment), but the trials follow people for a short time. A credit that assumes lifelong use is not supported by this evidence.
- **Persistence is the key unknown:** if many users stop within a year, the benefit may fade. The review above says side effects hurt persistence, but this brief did not find a persistence rate.
- **Claims and underwriting questions worth asking:** current or planned surgery or endoscopy, eye history, family or personal thyroid history, how long on the drug and at what dose.
- **Real-world data:** a 2025 study emulating a trial in adults with obesity and no diabetes looked at heart and kidney outcomes (Tang 2025). Its results were not read for this brief.

4. Sources (with identifiers)

1. Stefanou M-I et al. Ther Adv Neurol Disord 2024. DOI 10.1177/17562864241281903 · PMID 39345822
2. Zhou X-Y et al. Diabetol Metab Syndr 2026. DOI 10.1186/s13098-026-02163-7 · PMID 42316327
3. Zhang J et al. Metabolism 2026. DOI 10.1016/j.metabol.2025.156433 · PMID 41207615
4. Jalleh RJ et al. J Clin Invest 2026. DOI 10.1172/JCI194740 · PMID 41697736
5. Ueda P. J Intern Med 2026. DOI 10.1111/joim.70170 · PMID 42851139
6. Tang H et al. Diabetes Obes Metab 2025. DOI 10.1111/dom.70054 · PMID 40874398

5. What this brief is not

An evidence summary for discussion. It is not medical advice, an underwriting guideline, an actuarial assumption or legal advice. It was built from abstracts found through the Firecrawl research index and PubMed, not from full papers: the index returned no full text for three papers I tried. Numbers are quoted as the abstracts give them, not re-computed. Sections 1 and 2 report the papers; section 3 is my judgment. Prepared by Syed Shafai, syed.shafai@gmail.com. Abstract data from PubMed.